

The McGaugh Interpolation Function as a Poisson Survival Probability: An Observation and Its Implications for the External Field Effect

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Abstract

We note that the empirical MOND interpolation function $\mu(x) = 1 - \exp(-\sqrt{x})$, which provides the best fit to the Radial Acceleration Relation across 2,693 data points from 153 galaxies (McGaugh, Lelli & Schombert 2016), is mathematically identical to the Poisson survival probability $P(N \geq 1)$ with mean occupation number $\lambda = \sqrt{x}$, where $x = g_{\text{bar}}/g^*$. This identification, which to our knowledge has not previously appeared in the literature, has a non-trivial consequence: it provides a mechanism by which a linear gravitational field theory can produce the External Field Effect (EFE) without nonlinear field equations. The concavity of $\sqrt{x}$ in the Poisson activation rate ensures that an external gravitational field suppresses internal MOND dynamics, reproducing the observed EFE through quantum statistical nonlinearity rather than classical field nonlinearity. We derive the suppression factor, show it matches MOND+EFE predictions for strong external fields, and identify a discriminating prediction in the transition regime ($x_{\text{EFE}} = 2-4$) testable with JWST proper motion campaigns. We discuss possible physical interpretations, including de Sitter vacuum statistics, while noting that the derivation of $\lambda = \sqrt{x}$ from first principles remains an open problem.

Keywords: MOND, Radial Acceleration Relation, Poisson statistics, External Field Effect, interpolation function, modified gravity

1. The Observation

The Radial Acceleration Relation (RAR) established by McGaugh, Lelli & Schombert (2016) from 153 SPARC galaxies relates the observed gravitational acceleration g_{obs} to the baryonic Newtonian acceleration g_{bar} through an interpolation function μ :

$$g_{\text{obs}} = g_{\text{bar}} / \mu(g_{\text{bar}} / g^*)$$

where $g^* = 1.20 \times 10^{-10} \text{ m/s}^2$. The function providing the best fit to the data is:

$$\mu(x) = 1 - \exp(-\sqrt{x}) \quad [\text{McGaugh function}]$$

where $x = g_{\text{bar}}/g^*$. This function, which has appeared in the MOND literature prior to McGaugh et al. (Milgrom 1994; see also Milgrom 2016 for historical discussion), satisfies the required limits: $\mu \rightarrow 1$ for $x \gg 1$ (Newtonian regime) and $\mu \rightarrow \sqrt{x}$ for $x \ll 1$ (deep MOND regime). It provides a superior fit to galaxy kinematic data compared to the standard function $\mu_n = x/\sqrt{1+x^2}$ (Milgrom 1983) and the simple function $\mu_s = x/(1+x)$ (Famaey & Binney 2005).

We note the following identity. A Poisson random variable N with mean λ has survival probability:

$$P(N \geq 1) = 1 - P(N = 0) = 1 - \exp(-\lambda)$$

Setting $\lambda = \sqrt{x}$:

$$P(N \geq 1) = 1 - \exp(-\sqrt{x}) = \mu(x)$$

The McGaugh interpolation function is exactly the Poisson survival probability with mean occupation number $\lambda = \sqrt{x}$. This is an exact mathematical identity, not an approximation. To our knowledge, this connection has not previously been noted in the literature.

We emphasize what this observation is and is not. It IS a mathematical identity between the best-fit empirical MOND function and a fundamental probability distribution. It is NOT, by itself, a physical derivation of MOND. Any monotonically increasing function from 0 to 1 can be written as a survival probability $1 - \exp(-g(x))$ for some $g(x)$; what distinguishes the present case is that $g(x) = \sqrt{x}$ is the simplest algebraic function with the correct deep-MOND scaling, and that Poisson statistics have natural physical interpretations in quantum field theory.

2. Physical Interpretation of the Limits

The Poisson identification gives physical meaning to the two MOND regimes:

Newtonian regime ($x \gg 1$, high acceleration): $\lambda = \sqrt{x} \gg 1$. Many quanta are excited on average. The survival probability $P(N \geq 1) \rightarrow 1$ with near-certainty. The system behaves classically and $g_{\text{obs}} \rightarrow g_{\text{bar}}$. This is the classical field limit of any Poisson-activated process.

Deep MOND regime ($x \ll 1$, low acceleration): $\lambda = \sqrt{x} \ll 1$. Very few quanta are excited. The survival probability $P(N \geq 1) \rightarrow \lambda = \sqrt{x}$. The observed acceleration becomes $g_{\text{obs}} = g_{\text{bar}}/\sqrt{x} = \sqrt{g_{\text{bar}} * g^*}$, the deep MOND formula. This is the quantum vacuum limit where excitation is rare.

The transition ($x \sim 1$): $\lambda \sim 1$. The system transitions from rare excitation to near-certain excitation. The Poisson function $1 - \exp(-\sqrt{x})$ provides the smooth interpolation that the galaxy data require.

If the Poisson identification has physical content, it suggests that the MOND phenomenon is a transition between two regimes of a quantum field: from a vacuum state (few quanta, rare activation) to a classical state (many quanta, near-certain activation), with the transition governed by the acceleration scale g^* .

3. The External Field Effect from Poisson Concavity

3.1 The Problem

A key observational feature of MOND is the External Field Effect (EFE): an external gravitational field g_{ext} suppresses internal MOND dynamics when $g_{\text{ext}} > g_{\text{int}}$. This has been confirmed in globular cluster kinematics (Baumgardt & Hilker 2018; Haghi et al. 2019). In the standard MOND framework (AQUAL; Bekenstein & Milgrom 1984), the EFE arises from the nonlinearity of the modified Poisson equation. Linear field theories, in which gravitational potentials superpose, have generally been considered unable to produce the EFE. This has been a significant challenge for alternative MOND formulations based on linear field equations.

3.2 Resolution via Poisson Statistics

The Poisson identification resolves this challenge. Consider a test system with internal baryonic acceleration g_{int} embedded in an external field g_{ext} . In a linear field theory, the total baryonic acceleration at any point is:

$$g_{\text{total}} = g_{\text{int}} + g_{\text{ext}}$$

The Poisson activation rate is:

$$\lambda_{\text{total}} = \sqrt{g_{\text{total}} / g^*} = \sqrt{(g_{\text{int}} + g_{\text{ext}}) / g^*}$$

The function $f(g) = \sqrt{g}$ is concave ($f'(g) < 0$). By the concavity inequality:

$$\sqrt{g_{\text{int}} + g_{\text{ext}}} < \sqrt{g_{\text{int}}} + \sqrt{g_{\text{ext}}}$$

The external field consumes part of the activation budget. The marginal activation rate attributable to the internal field, when $g_{\text{int}} \ll g_{\text{ext}}$, is:

$$\begin{aligned} \Delta \lambda_{\text{int}} &= \lambda(g_{\text{int}} + g_{\text{ext}}) - \lambda(g_{\text{ext}}) \\ &= \sqrt{(g_{\text{int}} + g_{\text{ext}})/g^*} - \sqrt{g_{\text{ext}}/g^*} \\ &\sim g_{\text{int}} / (2 * \sqrt{g_{\text{ext}} * g^*}) \end{aligned}$$

Compared to the isolated case $\lambda_{\text{int,iso}} = \sqrt{g_{\text{int}}/g^*}$, the suppression ratio is:

$$\eta_{\text{EFE}} = \Delta \lambda_{\text{int}} / \lambda_{\text{int,iso}} = (1/2) * \sqrt{g_{\text{int}}/g_{\text{ext}}}$$

For $x_{\text{EFE}} = g_{\text{ext}}/g_{\text{int}} \gg 1$ (strong external field), $\eta_{\text{EFE}} \rightarrow 0$: internal MOND dynamics are suppressed toward Newtonian behavior. This is the External Field Effect, emerging from a linear field theory through the nonlinearity of the Poisson probability function.

The physical mechanism is fundamentally different from AQUAL: in AQUAL, the EFE arises from classical nonlinearity of the field equation. Here, it arises from quantum statistical nonlinearity of the activation probability applied to a linear field. Both mechanisms produce the same qualitative behavior.

3.3 Discriminating Prediction

The two mechanisms make quantitatively different predictions in the transition regime $x_{\text{EFE}} = 2$ -4. We parameterize the MOND fraction retained in the presence of an external field as $w(x_{\text{EFE}})$, where $w = 1$ means full MOND and $w = 0$ means Newtonian.

Table 1. Discriminating predictions at $x_{\text{EFE}} = g_{\text{ext}}/g_{\text{int}} = 2$.

Model	w at $x_{\text{EFE}} = 2$	Scaling	Source
MOND+EFE (Haghi)	20%	$\sim x^{-2}$	Field nonlinearity
Poisson-EFE	11%	$\sim x^{-1}$	Probability nonlinearity

The two models differ by approximately a factor of 2 in the MOND fraction at $x_{\text{EFE}} = 2$. This is testable with globular clusters at galactocentric distances of 5-15 kpc with mass below 10^4 solar masses, where x_{EFE} falls in the transition range. JWST proper motion campaigns can provide the velocity dispersion precision needed to distinguish the models.

For $x_{\text{EFE}} > 4$, both models predict nearly Newtonian behavior and are observationally indistinguishable.

4. Relation to Prior Work

4.1 Other Statistical Physics Approaches to MOND

Pazy (2013) derived MOND interpolation functions from quantum statistics of degrees of freedom on holographic screens, relating the MOND acceleration scale to the Fermi energy of a two-dimensional fermion gas. That work derives Fermi-Dirac and Bose-Einstein interpolation functions, which differ from the Poisson result presented here. The present work is complementary: both interpret the MOND transition as a quantum statistical phenomenon, but the specific statistics differ. Pazy's approach is more developed theoretically; the present approach yields the empirically superior McGaugh function.

Verlinde (2017) derives MOND from de Sitter entropy elasticity, obtaining the Milgrom interpolation function $\mu = x/\sqrt{1+x^2}$ rather than the McGaugh function. Both frameworks involve $\sqrt{x}$ scaling from de Sitter physics, but the statistical treatment differs. Verlinde's entropy approach and the Poisson approach presented here may be complementary descriptions of the same underlying phenomenon.

4.2 History of the McGaugh Function

The function $\mu(x) = 1 - \exp(-\sqrt{x})$ was not first proposed by McGaugh et al. (2016). As Milgrom (2016) notes, this function and its equivalent ν -form had appeared in the MOND literature prior to 2016. McGaugh et al.'s contribution was the definitive empirical demonstration, using the SPARC dataset, that this specific function provides the best fit to the RAR across 153 galaxies. We use the label 'McGaugh function' for convenience but acknowledge the historical precedent.

4.3 Other Interpolation Functions

The Milgrom standard function $\mu_n = x/\sqrt{1+x^2}$ and the Famaey-Binney simple function $\mu_s = x/(1+x)$ have no known Poisson interpretation. Both can be written as survival probabilities $1 - \exp(-g(x))$ for appropriate $g(x)$, but the resulting $g(x)$ lacks the simple algebraic form $\sqrt{x}$ that connects to dimensional analysis and de Sitter physics. The uniqueness of the

Poisson identification among standard MOND functions is a point in its favor, though it does not constitute proof of physical relevance.

5. Possible Physical Origins of $\lambda = \sqrt{x}$

The Poisson identification raises the question: why should the mean occupation number be $\lambda = \sqrt{g_{\text{bar}}/g^*}$? We note three observations, short of a derivation:

Dimensional constraint. Given only g_{bar} and g^* (both with dimensions m/s^2), the dimensionless combination $x = g_{\text{bar}}/g^*$ is unique. Among power laws $\lambda = x^\alpha$, only $\alpha = 1/2$ yields the correct deep-MOND formula $g_{\text{obs}} = \sqrt{g_{\text{bar}} * g^*}$. This fixes $\alpha = 1/2$ but does not explain why the occupation number should be a power law.

De Sitter temperature scale. In de Sitter space with cosmological constant Λ , the Bunch-Davies vacuum state has temperature $T_{\text{dS}} = \hbar * H_\Lambda / (2\pi k_B)$, which sets an acceleration scale $g_{\text{dS}} = c * H_\Lambda / (2\pi) = 1.04 \times 10^{-10} \text{ m/s}^2$. This is within 13% of the observed $g^* = 1.20 \times 10^{-10} \text{ m/s}^2$. The proximity suggests that g^* may originate from de Sitter thermodynamics. If so, the Poisson process may describe the excitation of quantum field modes in the Bunch-Davies thermal state by baryonic gravitational sources. The 13% discrepancy is within the range of systematic uncertainties in stellar mass-to-light ratios used in RAR calibration.

Convergence with Verlinde. Verlinde (2017) independently obtains $\sqrt{x}$ scaling from de Sitter entropy elasticity via a completely different route. The convergence of two independent de Sitter-based approaches on the same $\sqrt{x}$ structure suggests this scaling may be a genuine feature of gravity in de Sitter space rather than an artifact of either framework.

A rigorous derivation of $\lambda = \sqrt{x}$ from quantum field theory in de Sitter background remains an open problem. Such a derivation would require computing the mean occupation number of field modes excited by a baryonic source in the Bunch-Davies vacuum state and showing that the result depends only on the combination g_{bar}/g^* . This is equivalent to deriving MOND from first principles and is beyond the scope of the present work.

6. Necessary Conditions and Falsification

The Poisson interpretation requires specific conditions to hold:

Condition 1: The McGaugh function must remain the best empirical fit. The function belongs to a one-parameter family $\nu(z) = [1 - \exp(-z^{(\delta/2)})]^{(-1/\delta)}$ with δ as a free parameter. The Poisson interpretation requires $\delta = 1$ exactly. If future kinematic surveys (DESI, MIGHTEE-HI) with larger samples and higher precision prefer $\delta \neq 1$, the Poisson interpretation fails.

Condition 2: The EFE transition regime prediction must hold. At $x_{\text{EFE}} = 2-4$, Poisson-EFE predicts approximately 11% MOND character retention, while MOND+EFE (Haghi et al. 2019) predicts approximately 20%. If the data favor the MOND+EFE prediction, the Poisson-EFE mechanism is ruled out for EFE while the mathematical identity remains.

Condition 3: Deep-MOND scatter should be Poisson-distributed. If the Poisson interpretation is physical, the scatter in the RAR at low accelerations ($x \ll 1$) should follow Poisson statistics rather than Gaussian statistics. This is testable with high-precision kinematic surveys but has not yet been checked.

Condition 4: The acceleration scale gap must close. The de Sitter prediction $g_{\text{dS}} = 1.04 \times 10^{-10} \text{ m/s}^2$ is 13% below the observed $g^* = 1.20 \times 10^{-10} \text{ m/s}^2$. JWST near-IR stellar population synthesis on SPARC galaxies will provide improved stellar mass-to-light ratios. If the revised mass-to-light ratios do not close this gap, the de Sitter origin of g^* is disfavored.

7. Conclusions

We have identified an exact mathematical connection between the empirically best-fit MOND interpolation function $\mu(x) = 1 - \exp(-\sqrt{x})$ and the Poisson survival probability $P(N \geq 1)$ with mean occupation $\lambda = \sqrt{x}$. This observation, which to our knowledge is new, has the following implications:

(1) It provides a statistical physics framing of MOND: the transition from Newtonian to modified dynamics corresponds to the transition from rare to near-certain Poisson excitation of a quantum process with rate $\lambda = \sqrt{g_{\text{bar}}/g^*}$.

(2) It resolves the External Field Effect problem for linear field theories. The concavity of $\sqrt{x}$ in the Poisson activation rate ensures that external fields suppress internal MOND dynamics without requiring nonlinear field equations. This mechanism produces a testable discriminating prediction in the transition regime $x_{\text{EFE}} = 2-4$.

(3) It motivates a specific open problem in quantum field theory: derive $\lambda = \sqrt{x}$ from the Bunch-Davies vacuum state of de Sitter space. A successful derivation would connect the MOND acceleration scale, the transition shape, and the deep-MOND formula to de Sitter thermodynamics.

We do not claim to have derived MOND from first principles. The Poisson identification is an observation whose physical content depends on whether $\lambda = \sqrt{x}$ can be derived from an underlying theory. What we have shown is that the best empirical fit to galaxy kinematics is not an arbitrary function but a fundamental probability distribution, and that this identification has a concrete, testable consequence for the External Field Effect.

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